Yuexi Du

Biomedical Engineering :: Multimodal AI :: Medical Image Analysis
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EDUCATION

Yale University, New Haven, U.S. *Sept. 2022 – Present*

- **Ph.D. Candidate**, Biomedical Engineering, Medical Image Analysis
- Advised by Prof. Nicha C. Dvornek.

University of Michigan, Ann Arbor, U.S. *Sept. 2020 – May 2022*

- **B.S.**, Computer Science; Mathematics Minor; **GPA: 3.97/4.00**
- Coursework: Computer Organization (A+), Computer Vision (A+), Advanced Topics in CV (A+)

Shanghai Jiao Tong University, Shanghai, China *Sept. 2018 – Aug. 2022*

- **B.S.**, Electrical and Computer Engineering; **GPA: 3.70/4.00 (Top 10%)**
- Coursework: Programming and Data Structures (A+), Honors Mathematics IV (A+)

SELECTED PUBLICATIONS

[1] **Yuexi Du**, Leya Barrientos, Laura Sheiman, John Lewin, Hemant D. Tagare, Nicha C. Dvornek. “MammoFlow: Multiview Mammogram Synthesis with Anatomically Consistent Flow Matching.” *MICCAI 2026*. [Paper] [Code]

[2] **Yuexi Du**, Jinglu Wang, Shujie Liu, Nicha C. Dvornek, Yan Lu. “CARE: Towards Clinical Accountability in Multi-Modal Medical Reasoning with an Evidence-Grounded Agentic Framework.” *ICLR 2026*. [Paper] [Project]

[3] **Yuexi Du**, Lihui Chen, Nicha C. Dvornek. “Geometry-Guided Local Alignment for Multi-View Visual Language Pre-Training in Mammography.” *MICCAI 2025*. [Paper][Code]

[4] **Yuexi Du**, Jiazhen Zhang, Nicha C. Dvornek, John A. Onofrey. “GMR-Conv: An Efficient Rotation and Reflection Equivariant Convolution Kernel Using Gaussian Mixture Rings.” *arXiv, 2025*. [Paper] [Code]

[5] **Yuexi Du**, John A. Onofrey, Nicha C. Dvornek. “Multi-View and Multi-Scale Alignment for Contrastive Language-Image Pre-training in Mammography.” *IPMI 2025*, Best Paper Runner-up, Oral. [Paper] [Code]

[6] **Yuexi Du**, Jiazhen Zhang, Tal Zeevi, Nicha C. Dvornek, John A. Onofrey. “SRE-Conv: Symmetric Rotation Equivariant Convolution for Biomedical Image Classification.” *IEEE ISBI 2025*. [Paper] [Code]

[7] **Yuexi Du**, Brian Chang, Nicha C. Dvornek. “CLEFT: Language-Image Contrastive Learning with Efficient Large Language Model and Prompt Fine-Tuning.” *MICCAI 2024*. [Paper] [Code]

[8] **Yuexi Du**, Regina J. Hooley, John Lewin, Nicha C. Dvornek. “SIFT-DBT: Self-supervised Initialization and Fine-Tuning for Imbalanced Digital Breast Tomosynthesis Image Classification.” *IEEE ISBI 2024*. [Paper] [Code]

[9] **Yuexi Du**, Ziyang Chen, Justin Salamon, Bryan Russell, Andrew Owens. “Conditional Generation of Audio from Video via Foley Analogies.” *CVPR 2023*. [Paper] [Code]

[10] Xiyue Wang, **Yuexi Du**, Sen Yang, Jun Zhang, Minghui Wang, Jing Zhang, Wei Yang, Junzhou Huang, Xiao Han. “RetCCL: Clustering-Guided Contrastive Learning for Whole-Slide Image Retrieval.” *Medical Image Analysis 83 (2023): 102645*. [Paper] [Code]

INDUSTRY EXPERIENCE

Camera Motion Autorater & E2E Video Generation Refinement *Mentored by Dr. Yan Xu*
GenMedia @ Google Cloud *May 2026 – Present*

- Developed an agentic VLM autorater with RL fine-tuning on Gemini to identify camera motion from temporal visual cues, including optical flow; improved accuracy by more than 20% over SOTA MLLMs.
- Integrated the tuned autorater into an end-to-end, training-free refinement pipeline for Veo and Omni camera controls, improving prompt–motion alignment by over 15%.

Visual Evidence-Grounded Medical VQA Agents *Mentored by Dr. Jinglu Wang*
Media Computing Group @ Microsoft Research Asia *June 2025 – Sept. 2025*

- Developed CARE, an evidence-grounded medical-VQA framework combining entity proposal, expert segmentation, grounded reasoning, and coordinator-based planning and review.
- Fine-tuned the agentic framework with RLVR, improving average accuracy by 10.9% over a same-size 10B SOTA model and 5.2% over a pretrained 32B baseline.

Content-Based Pathology Image Retrieval *Mentored by Dr. Xiao Han & Sen Yang*
AI Healthcare Group @ Tencent AI Lab *May 2021 – Sept. 2021*

- Designed a clustering-guided contrastive-learning method and an efficient content-based retrieval pipeline for whole-slide pathology images.
- Improved retrieval performance on TCGA by over 10% and downstream lung-cancer patch classification by over 5% versus baseline methods.

RESEARCH EXPERIENCE

Accountable Multimodal Medical AI & Generative Modeling — *Advised by Nicha C. Dvornek*
IPAG @ Yale University — *June 2023 – Present*

- Developed CARE, a clinically accountable multimodal framework with evidence-grounded agentic reasoning and explicit ROI support; improved average accuracy by 10.9% over a same-size SOTA model.
- Designed MammoFlow with geometry-guided cross-view flow matching for anatomically consistent paired mammogram synthesis; passed radiologist evaluation and improved downstream classification AUC by

Rotation- & Reflection-Equivariant Deep Learning — *Advised by John A. Onofrey*
IPAG @ Yale University — *Sept. 2022 – Present*

- Developed SRE-Conv and GMR-Conv, efficient convolution kernels that encode radial symmetry while reducing discretization error and computational overhead.
- Validated GMR-Conv on eight classification and one segmentation datasets; released open-source code and a PyPI package.

Controlled & Self-Supervised Diffusion MRI Denoising — *Advised by Andre Wibisono*
Yale University — *Jan. 2023 – May 2023*

- Added controlled frozen encoders/decoders and zero-convolution to an attention-based diffusion U-Net; used DDIM to stabilize denoising across real-world diffusion-MRI datasets.
- Improved denoised-image SSIM by over 10% on both datasets and strengthened downstream CSD and DTI modeling performance.

TEACHING EXPERIENCE

ENAS 912: Biomedical Image Processing, TF, Yale; Profs. James Duncan & Lawrence Staib — *Sept. 2023 – Dec. 2023*
Led office hours and review sessions; graded assignments.
EECS 442: Computer Vision, Instructional Aide, UMich; Prof. David Fouhey — *Jan. 2022 – Apr. 2022*
Led office hours; designed assignments/projects; managed Piazza.
VR 246: Intro. to Comics & Graphic Novels, TA, SJTU; Prof. Joelle Tybon — *May 2021 – Aug. 2021*
Led office hours and graded assignments.
VE 101: Intro. to Computer & Programming, TA, SJTU; Prof. Jigang Wu — *Aug. 2020 – Dec. 2020*
Led review classes and office hours; graded work and designed lab questions.

SELECTED AWARDS & HONORS

NIH Registration Grant, MICCAI — *2026*
Star of Tomorrow Internship Program, Microsoft Research Asia — *2025*
Second Place, François Erbsmann Prize for Best Paper, IPMI — *2025*
CXR-LT Challenge Overall Best Solution, MICCAI — *2024*
Conference Travel Funding, Yale University — *2024*
University Fellowship, Yale University — *2022–2024*
Outstanding Graduate of Shanghai, Shanghai Jiao Tong University — *2022*
James B. Angell Scholar, University of Michigan — *2022*
Dean's Honor List & University Honor, University of Michigan — *2020–2022*
Undergraduate Scholarship of Excellence, Shanghai Jiao Tong University — *2019–2020*

SERVICES

Conference & Journal Reviewer, New Haven, U.S. — *Dec. 2023 – Present*
CVPR, ICCV, ECCV, ICML, NeurIPS, ICLR, AAAI, AISTATS, IJCAI, MICCAI, ISBI; IEEE TPAMI, IEEE TMM, IEEE TMI, IJCV, and *Medical Image Analysis*.
Leader, Organization Department, Joint Institute Student Union, SJTU — *May 2019 – June 2020*
Leader, Volunteer Teaching Group, Yunnan Kuang Chang Primary School — *Dec. 2019 – Feb. 2020*

SKILLS

Programming: Python, C/C++, MATLAB, Bash, R, HTML, JavaScript, Go, LaTeX.
ML / Systems: PyTorch, PyTorch Lightning, Hugging Face Transformers, vLLM, verl, ms-swift, OpenRLHF, Ray, DeepSpeed, Triton, FAISS.